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Assignment 6

For our senior design project, we created an application that enables a competitive coding environment. My contributions primarily focused on the backend, specifically implementing the 1v1 and 2v2 multiplayer functionality. In addition to UI development and general programming, this required applying and expanding upon networking concepts I learned in class. I utilized Supabase Realtime to configure WebSockets for seamless real-time communication. Beyond technical skills, this experience also improved my project management and communication abilities.

A major success was successfully implementing WebSockets to create an engaging and enjoyable game loop. I also played a key role in developing the practice page, designing it to be modern and dynamic, with robust filtering options for topic and difficulty. One significant challenge we faced was conceptualizing the 2v2 competitive mode. We needed to ensure it offered a distinct experience from the 1v1 mode. We ultimately designed a team-based format where two users are graded on their combined completion rate, introducing an exciting layer of strategy and reliance on teamwork.

As a team, we successfully met all our project requirements. We delivered core features, including the 1v1 and 2v2 modes, a practice page, a friends list, community features, a leaderboard, notifications, and a history page. Through this collaborative process, we learned that effective teamwork significantly boosted our productivity. However, we did face challenges; reaching a consensus on the product's scope and final vision was occasionally difficult due to conflicting ideas.

Our team functioned with an equal distribution of effort, with everyone contributing their unique strengths to different areas of the application. All of my teammates deserve recognition for their critical roles in bringing this project to life. Overall, building this application was a highly rewarding capstone experience. Applying our diverse skills to a comprehensive project taught me valuable lessons about my own personal and professional strengths and weaknesses, ultimately preparing me to be a better professional in my career.